

Cambridge (CIE) IGCSE Additional Maths



Your notes

Straight-Line Graphs

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Linear Graphs



Your notes

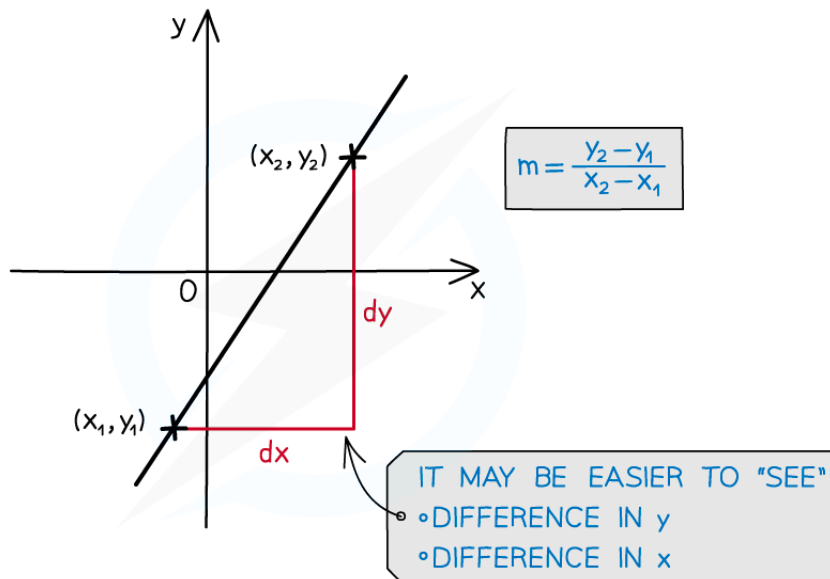
Equations of a Straight Line

How do I find the gradient of a straight line?

- Find two points that the line passes through with coordinates (x_1, y_1) and (x_2, y_2)
- The gradient between these two points is calculated by

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- The gradient of a straight line measures its **slope**
 - A line with gradient 1 will go up 1 unit for every unit it goes to the right
 - A line with gradient -2 will go down two units for every unit it goes to the right



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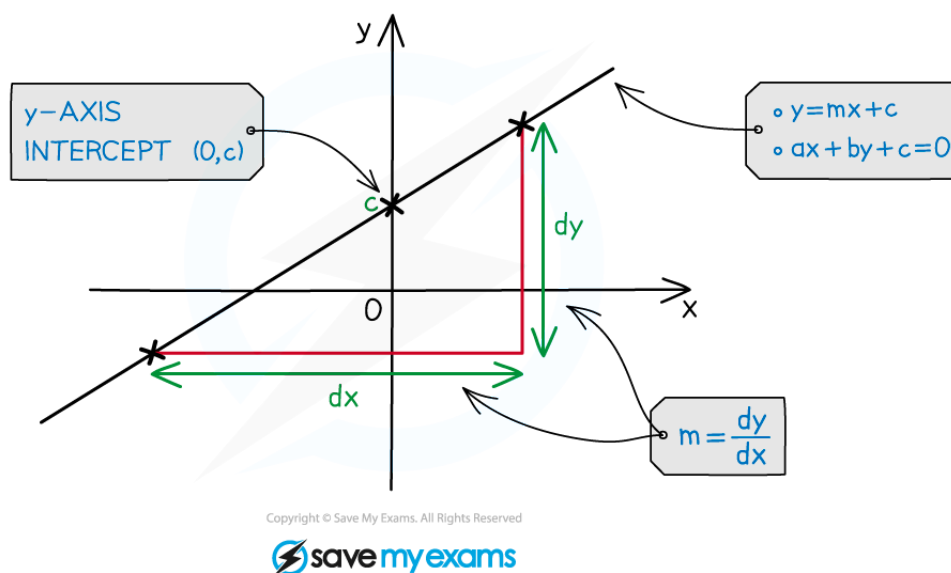


What are the equations of a straight line?



Your notes

- $y = mx + c$
 - This is the **gradient-intercept form**
 - It clearly shows the gradient m and the y -intercept $(0, c)$
- $y - y_1 = m(x - x_1)$
- This is the **point-gradient form**
 - It clearly shows the gradient m and a point on the line (x_1, y_1)
- $ax + by + d = 0$
 - This is the **general form**
 - You can quickly get the x -intercept $\left(-\frac{d}{a}, 0\right)$ and y -intercept $\left(0, -\frac{d}{b}\right)$



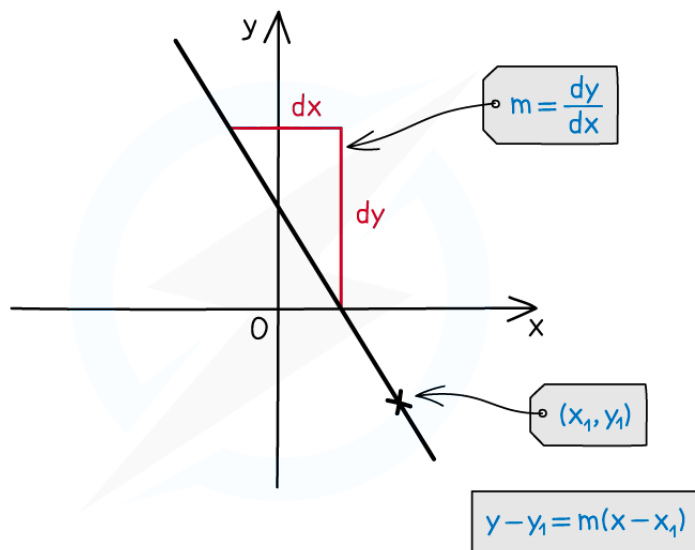
How do I find an equation of a straight line?

- You will need the gradient
 - If you are given two points then first find the gradient
- It is easiest to start with the **point-gradient form**

- then rearrange into whatever form is required
 - multiplying both sides by any denominators will get rid of fractions



Your notes



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Examiner Tips and Tricks

- Quickly sketching the graph of the straight line(s) can be helpful if you are struggling with an exam question
- Ensure you state equations of straight lines in the format required
 - Usually $y = mx + c$ or $ax + by + d = 0$
 - Check whether coefficients need to be integers (they usually are for $ax + by + d = 0$)



Worked Example

(a) Find the equation of the straight line with gradient 3 that passes through (5, 4).

We know that the gradient is 3 so the line takes the form



Your notes

$$y = 3x + c$$

To find the value of c , substitute $(5, 4)$ into the equation

$$4 = 3(5) + c$$

$$4 = 15 + c$$

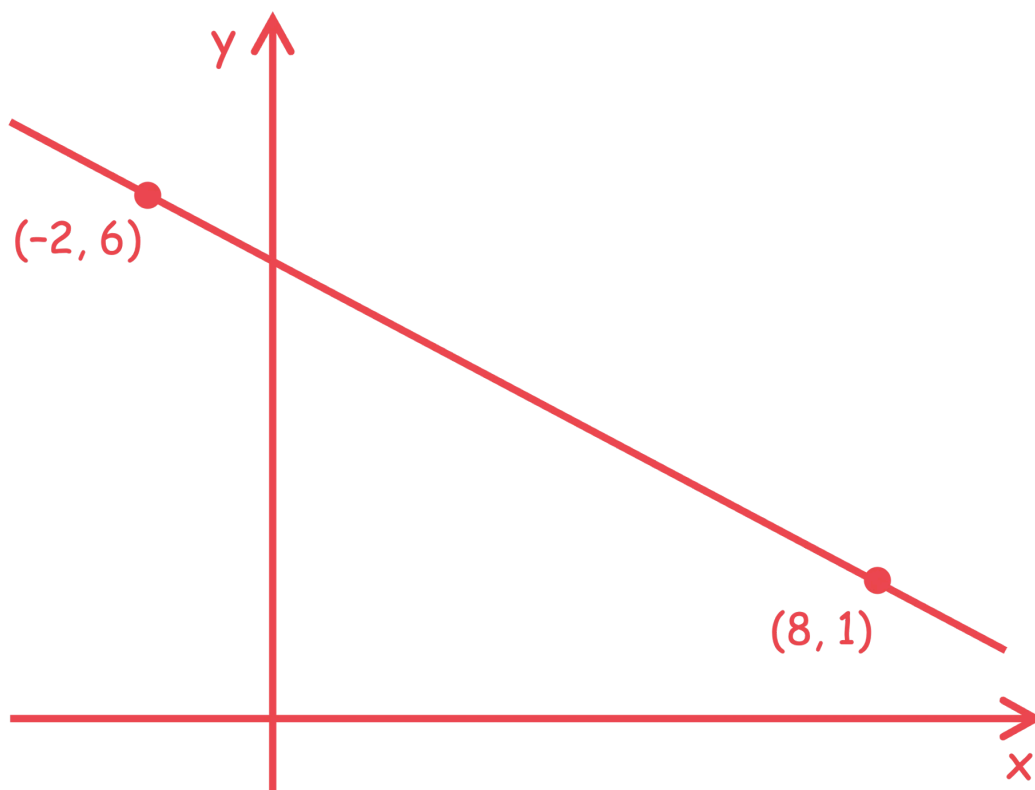
$$c = -11$$

Replace c with -11 to complete the equation of the line

$$y = 3x - 11$$

(b) Find the equation of the straight line that passes through $(-2, 6)$ and $(8, 1)$.

You may find it helpful to sketch the information given





Your notes

First find m , the gradient

$$\begin{aligned} m &= \frac{6-1}{-2-8} \\ &= \frac{5}{-10} \\ &= -\frac{1}{2} \end{aligned}$$

We know that the line takes the form

$$y = -\frac{1}{2}x + c$$

To find the value of c , substitute either of the given points into this equation. Here we will pick $(8, 1)$ as it doesn't contain negative numbers so is easier to work with

$$\begin{aligned} 1 &= -\frac{1}{2}(8) + c \\ 1 &= -4 + c \\ c &= 5 \end{aligned}$$

Replace c with -11 to complete the equation of the line

$$y = -\frac{1}{2}x + 5$$

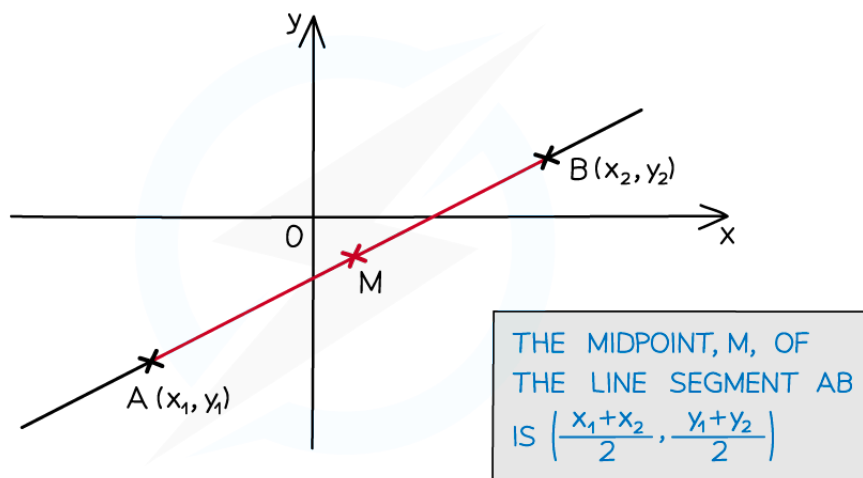
We can check against our sketch that this equation looks correct- it has a negative gradient and it crosses the y -axis between 1 and 6

Midpoint of a Straight Line

How do I find the midpoint of a line (segment)?

- The midpoint of a line will be the same distance from both endpoints
- You can think of a midpoint as being the average (mean) of two coordinates
- The midpoint of (x_1, y_1) and (x_2, y_2) is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$



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- M is often used as the midpoint between two points (x_1, y_1) (x_2, y_2)
- It is the average of both the **x** and **y** coordinates



Worked Example

The coordinates of A are $(-4, 3)$ and the coordinates of B are $(8, -12)$.

Find M, the midpoint of AB.

The midpoint can be found using $M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

Fill in the values of x and y from each coordinate

$$\left(\frac{-4 + 8}{2}, \frac{3 + -12}{2} \right) = \left(\frac{4}{2}, \frac{-9}{2} \right)$$

Simplify

$M = (2, -4.5)$

 Your notes

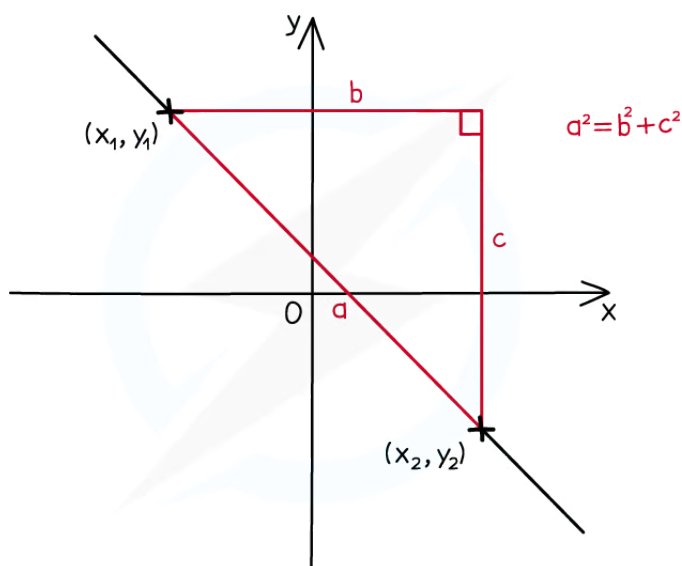
Length of a Straight Line

How do I calculate the length of a line?

- The distance between two points with coordinates (x_1, y_1) and (x_2, y_2) can be found using the formula

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- This formula is really just Pythagoras' Theorem $a^2 = b^2 + c^2$, applied to the difference in the x -coordinates and the difference in the y -coordinates;



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- You may be asked to find the length of a diagonal in 3D space
 - This can be answered using **3D Pythagoras**





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Examiner Tips and Tricks

- Work with the **square** of a distance for as long as possible as this avoids early rounding errors
 - In the non-calculator paper your answer may need to be left as a surd
- Only square root when forced to or for a final answer, and use the ANS button (and other memory features) on your calculator



Worked Example

Point A has coordinates (3, -4) and point B has coordinates (-5, 2).

Calculate the distance of the line segment AB.

Using the formula for the distance between two points, $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

Substituting in the two given coordinates:

$$d = \sqrt{(3 - -5)^2 + (-4 - 2)^2}$$

Simplify:

$$d = \sqrt{(8)^2 + (-6)^2} = \sqrt{64 + 36} = \sqrt{100} = 10$$

Answer = 10 units



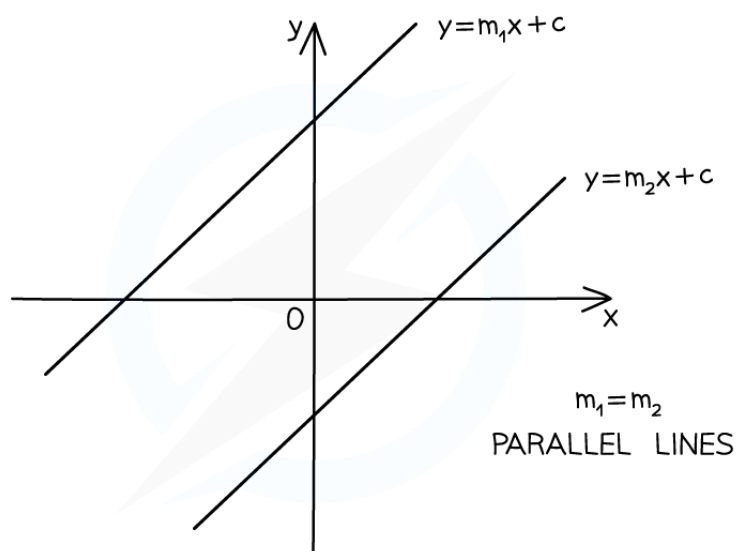
Your notes

Parallel & Perpendicular Lines

Parallel Lines

What are parallel lines?

- **Parallel lines** are always equidistant meaning they never intersect
- Parallel lines have the same gradient
 - If the gradient of line l_1 is m_1 and gradient of line l_2 is m_2 then...
 - $m_1 = m_2 \Rightarrow l_1 \text{ \& } l_2$ are parallel
 - $l_1 \text{ \& } l_2$ are parallel $\Rightarrow m_1 = m_2$
- To determine if two lines are parallel:
 - Rearrange into the gradient-intercept form $y = mx + c$
 - Compare the coefficients of x
 - If they are equal then the lines are parallel



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Your notes



Examiner Tips and Tricks

- Look for hidden parallel lines in an exam question
 - **Parallel lines** could be implied in an exam by phrases like "... at the same rate ..."
 - Check the properties of a geometrical shape



Worked Example

Find the equation of the line that is parallel to $y = 3x + 7$ and passes through (2,1).

As the gradient is the same, the line that is parallel will be in the form:

$$y = 3x + d$$

Substitute in the coordinate that the line passes through:

$$1 = 3(2) + d$$

Simplify:

$$1 = 6 + d$$

Subtract 6 from both sides:

$$-5 = d$$

Final answer:

$$y = 3x - 5$$

Perpendicular Lines

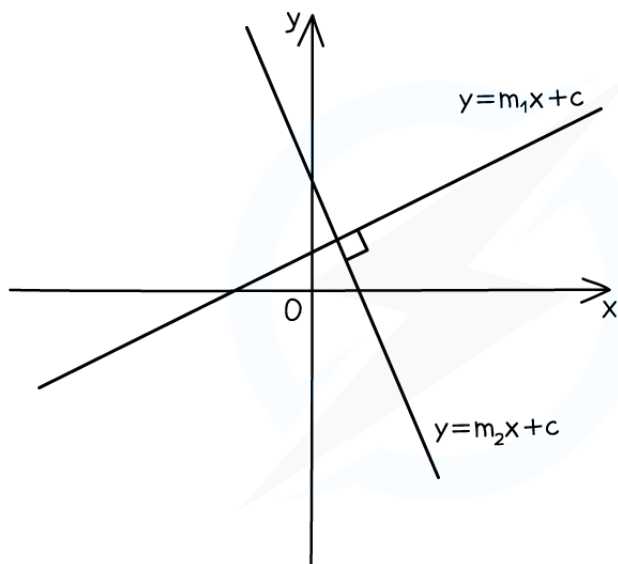
What are perpendicular lines?

- **Perpendicular lines** intersect at right angles
- The gradients of two perpendicular lines are negative reciprocals



Your notes

- If the gradient of line l_1 is m_1 and gradient of line l_2 is m_2 then...
 - $m_1 \times m_2 = -1 \Rightarrow l_1 \text{ \& } l_2$ are perpendicular
 - $l_1 \text{ \& } l_2$ are perpendicular $\Rightarrow m_1 \times m_2 = -1$
- To determine if two lines are perpendicular:
 - Rearrange into the gradient-intercept form $y = mx + c$
 - Compare the coefficients of x
 - If their product is -1 then they are perpendicular
- Be careful with horizontal and vertical lines
 - $x = p$ and $y = q$ are perpendicular where p and q are constants



$m_1 \times m_2 = -1$
PERPENDICULAR LINES

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Examiner Tips and Tricks

- Exam questions are good at “hiding” perpendicular lines



Your notes

- For example a **tangent** and a **radius** are **perpendicular**



Worked Example

Find the equation of the line that is perpendicular to $y = 2x - 2$ and passes through $(2, -3)$.

Leave your answer in the form $ax + by + c = 0$ where a, b, c are integers.

L is in the form $y = mx + c$ so we can see that its gradient is 2

$$m_1 = 2$$

Therefore the gradient of the line perpendicular to L will be the negative reciprocal of 2

$$m_2 = -\frac{1}{2}$$

Now we need to find c for the line we're after. Do this by substituting the point $(2, -3)$ into the equation $y = -\frac{1}{2}x + c$ and solving for c

$$-3 = -\frac{1}{2} \times 2 + c$$

$$-3 = -1 + c$$

$$c = -2$$

Now we know the line we want is

$$y = -\frac{1}{2}x - 2$$

But this is not in the form asked for in the question. So rearrange into the form $ax + by + c = 0$ where a, b and c are integers

$$y + \frac{1}{2}x + 2 = 0$$

$$2y + x + 4 = 0$$

Write the final answer

$$x + 2y + 4 = 0$$



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